Lesson 8-5 · Session 1

Graphing Inequalities

Name _____ Date _____ Period _____

Maryland Standard 6.EE.B.8 · Learning target — I can graph an inequality on a number line using an open or a closed dot.
Show a model, calculation, or explanation. Answer every part and label units when needed.

Exercises 1–12 earn 1 point for a correct response and 1 for supporting work. Spiral Review and Test Practice earn 1 point each.

Learn from a model

REMEMBER

- When graphing an inequality, shade in the direction of all the solutions (the arrow shows it continues forever).
- Always plot the boundary point first (open or closed), then shade the correct direction.
- Watch out — the dot type is decided by the symbol, not by the sign of the number.
- Check it — test one value on each side of the boundary to confirm the arrow direction.

WORKED EXAMPLE

Graphing an Inequality with a Closed Dot A delivery van is safe when its load is at most 900 pounds. Graph the loads that are safe.

Step 1 Write the inequality: let L = the pounds loaded, so L ≤ 900.

Step 2 Mark the boundary point at 900 with a CLOSED dot, because 900 is allowed.

Step 3 Solutions are 900 and everything smaller, so draw the arrow pointing left.

Step 4 Test a point under the arrow: 875 is allowed, which confirms the direction.

Answer L ≤ 900: a closed dot at 900 with an arrow pointing left.

Practice

1. Graph x ≤ 6 on a number line.

Graph. Mark the boundary with an open or a closed dot, then shade the arrow.

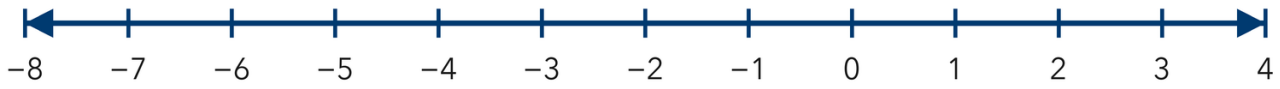


Answer _____

SHOW YOUR THINKING

2. Graph $x > -3$ on a number line.

Graph. Mark the boundary with an open or a closed dot, then shade the arrow.



Answer _____

SHOW YOUR THINKING

3. Write an inequality to match a graph with a closed circle at −1, shaded to the left.

Answer _____

SHOW YOUR THINKING

4. Write the direction the arrow points for each graph: left or right.

a. $x \geq 14$ _____
b. $n < 5$ _____
c. $p > -8$ _____
d. $k \leq 0$ _____
e. $w \geq -7$ _____

Answer _____

SHOW YOUR THINKING

5. Rock Wall. A rock wall lets a climber start only when the climber is at least 9 years old. Write the inequality using a and describe its graph.

Answer _____

RECORD YOUR RESPONSE

6. Write an inequality to match a graph with an open dot at 7 and an arrow pointing left.

Answer _____

SHOW YOUR THINKING

7. Cargo Elevator. A cargo elevator is safe with a load of at most 1,500 pounds. Write the inequality using L, draw a number line from 1,300 to 1,700, and graph the safe loads.

Graph. Mark the boundary with an open or a closed dot, then shade the arrow.



Answer _____

SHOW YOUR THINKING

8. Terrence graphed $m > 45$. He drew an open dot at 45 and then pointed his arrow to the left. Explain which part of his graph is wrong and why, then describe the corrected graph.

Answer _____

EXPLAIN WITH MATHEMATICAL EVIDENCE

9. Describe the graph of $n \leq 7$.

Answer _____ the graph

RECORD YOUR RESPONSE

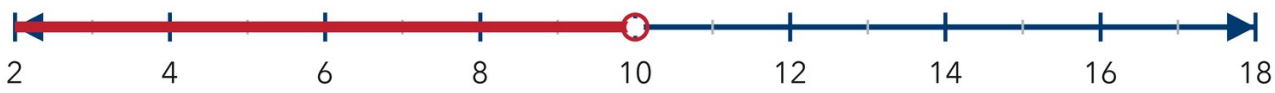
10. Describe the graph of $x > -2$.

Answer _____ the graph

RECORD YOUR RESPONSE

11. Read a Graph. A number line shows an open dot at 10 with an arrow pointing left. Write the inequality and say whether 10 is a solution.

Read the graph. This is the number line the problem describes.



Answer _____ the inequality and your answer

RECORD YOUR RESPONSE

12. Find the Mistake. Pat graphed $n \geq 6$ with an open dot. Explain the mistake.

Answer _____ the mistake

EXPLAIN WITH MATHEMATICAL EVIDENCE

Spiral review

13. From Lesson 8-4 Session 1. Write an inequality for "at least 100."

Answer _____

SHOW YOUR THINKING

14. From Lesson 8-3 Session 1. Solve $5m = 65$.

Answer _____

SHOW YOUR THINKING

15. From Lesson 6-7 Session 1. Find the GCF of 45 and 60.

Answer _____

SHOW YOUR THINKING

Test practice

16. The graph of $x \leq -4$ uses a —

□ A. square	□ B. open dot
□ C. closed dot	□ D. cross

17. For $n < 1$, the arrow points —

□ A. nowhere	□ B. right
□ C. both ways	□ D. left

18. A closed dot means the boundary value is —

□ A. a solution
□ B. not a solution
□ C. negative
□ D. zero

Check and revise

CHECK YOUR MATHEMATICS	HOW DID IT GO?
Use a filled point when the endpoint is included and an open point when it is excluded. Every part answered / Work shown / Units labeled when needed	I can do this: Independently / With a hint / With help Question to revisit: _____

One correction or question I want to discuss: _____

Test one proposed solution in the original equation or inequality. Show whether the statement is true.